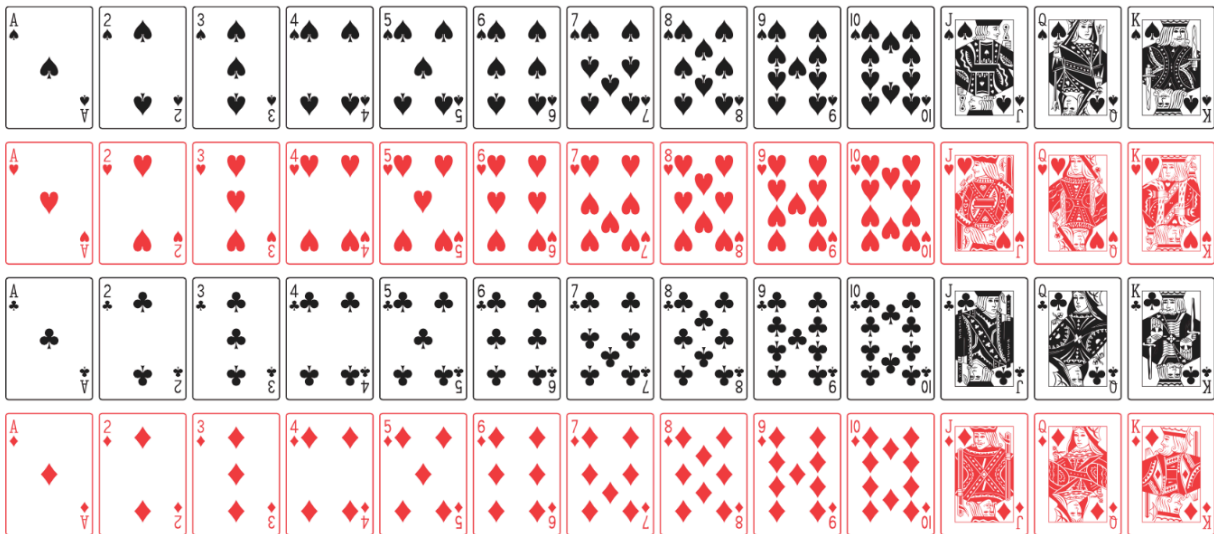


NAME:

STUDENT ID:

Consider a standard deck of 52 cards, as displayed in the following figure. In the first and third rows all cards are black; in the second and fourth rows, all cards are red.



Consider the following events when a card is randomly selected.

- A : card selected is a heart.
- B : card selected is the color black.
- C : card selected is a face card (i.e. J, Q, K)
- D : card selected is a King.

1. Find $P(A)$.

2. Find $P(C)$.

3. Find $P(D^C)$.

4. Find $P(A \cup B)$.

5. Create a box model to represent drawing a 3 or an 8 in a single draw from the deck of cards described above.

Consider the seven sided die (with sides 0, 1, 2, 3, 4, 5 and 6) from your reading questions. You roll this die once. Let:

- A be the event that a given roll yields an even number.
- B be the event that a given roll is greater than or equal to three.
- C be the event that the number appears in the phrase "Stat 20".

6. Find $P(A)$, $P(B)$ and $P(C)$. (**Hint:** write out the events *explicitly*. That will help with all three questions.)

7. Find $P(B \cap C)$

8. Find $P(A \cap B^C)$.

One number will be drawn at random from each of the two sets below:

$$A = \{1, 2, 3\}; B = \{1, 2, 3, 4\}$$

9. What is the probability the number drawn from A is greater than the one drawn from B ? (**Hint:** Write out the outcomes, for example $(1, 1)$, $(1, 2)$,)

10. What is the probability that the number drawn from A is equal to the one drawn from B ?

Consider an outcome space Ω with events A, B that are *not* mutually exclusive. Let $P(A) = 0.5$, $P(B) = 0.7$ and $P(A \cap B) = .4$. (Is it even *possible* for A and B to be mutually exclusive?)

11. Draw a Venn Diagram containing two circles, one representing A and the other B .

12. Shade in the space in the diagram above corresponding to event A .

13. Shade in the space in the diagram above corresponding to event B in a way that is different than the way you shaded in the space for event A .

14. Based on your shading, calculate the probability $P(A \cup B)$. Explain why you cannot add $P(A)$ and $P(B)$ directly to obtain this answer.